

National University of Modern Languages, Islamabad

Speech Processing Lab Report

Lab 7: MFCC Extraction and Neural Network Classification

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Objective

The purpose of this lab is to:

- Read and preprocess speech signals.
- Extract MFCC features (or fallback: mean & standard deviation).
- Train two types of neural networks:
 - Pattern Recognition Neural Network (PatternNet)
 - Feedforward Neural Network (FeedforwardNet)
- Evaluate classification accuracy using a custom function.

MATLAB Code: speech_lab.m

```
clc; clear; close all;

%% Step 1: Read the speech signals
disp('Reading speech files...');

try
    [x1, fs1] = audioread('D:\\matlab-jd\\speech1.wav');
    [x2, fs2] = audioread('D:\\matlab-jd\\speech2.wav');
catch ME
    error('Error reading audio files: %s', ME.message);
end

% Fix sampling frequency mismatch
if fs1 ~= fs2
    disp('Sampling frequencies do not match | resampling second signal...');
    x2 = resample(x2, fs1, fs2);
    fs2 = fs1;
end

%% Step 2: Feature Extraction (MFCC)
disp('Extracting MFCC features...');

numCoeffs = 13;
frameLength = round(0.025 * fs1);
frameOverlap = round(0.010 * fs1);
```

```

if exist('mfcc', 'file')
    coeff1 = mfcc(x1, fs1, 'WindowLength', frameLength, ...
        'OverlapLength', frameOverlap, 'NumCoeffs', numCoeffs);
    coeff2 = mfcc(x2, fs2, 'WindowLength', frameLength, ...
        'OverlapLength', frameOverlap, 'NumCoeffs', numCoeffs);
else
    warning('MFCC not found. Using mean & std instead.');
```

```

    coeff1 = [mean(x1), std(x1)];
    coeff2 = [mean(x2), std(x2)];
end

% Reduce MFCC to one feature vector per signal
feat1 = mean(coeff1, 1);
feat2 = mean(coeff2, 1);

% Dataset (features × samples)
X = [feat1; feat2]';
T = [1 0; 0 1]';

%% Step 3: Pattern Recognition NN
disp('Training PatternNet...');
net1 = patternnet(10);
net1.trainParam.showWindow = false;
net1 = train(net1, X, T);
YTest1 = net1(X);

%% Step 4: Feedforward Neural Network
disp('Training FeedforwardNet...');
net2 = feedforwardnet([15 10]);
net2.trainParam.showWindow = false;
net2 = train(net2, X, T);
YTest2 = net2(X);

%% Step 5: Evaluation
disp('Evaluating PatternNet:');
evaluate_performance(T, YTest1);

disp('Evaluating FeedforwardNet:');
evaluate_performance(T, YTest2);

```

MATLAB Code: `evaluate_performance.m`

```
function evaluate_performance(TTrue, YPred)

[~, pred] = max(YPred, [], 1);
[~, trueClass] = max(TTrue, [], 1);

figure;
plotconfusion(TTrue, YPred);
title('Confusion Matrix');

TP = sum(pred == 1 & trueClass == 1);
TN = sum(pred == 2 & trueClass == 2);
FP = sum(pred == 1 & trueClass == 2);
FN = sum(pred == 2 & trueClass == 1);

accuracy = (TP + TN) / (TP + TN + FP + FN);
precision = TP / (TP + FP + eps);
recall = TP / (TP + FN + eps);
f1 = 2 * (precision * recall) ...
    / (precision + recall + eps);

fprintf('Accuracy   : %.2f%%\n', accuracy * 100);
fprintf('Precision  : %.2f%%\n', precision * 100);
fprintf('Recall     : %.2f%%\n', recall * 100);
fprintf('F1 Score   : %.2f%%\n', f1 * 100);

end
```

Explanation

- Two speech files are loaded using `audioread()`.
- If sampling rates differ, the second signal is resampled.
- MFCC features are extracted using 25ms frames and 10ms overlap.
- If MFCC is unavailable, mean and standard deviation features are used.
- PatternNet and FeedforwardNet are trained using two classes.
- A custom function computes:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Output Images

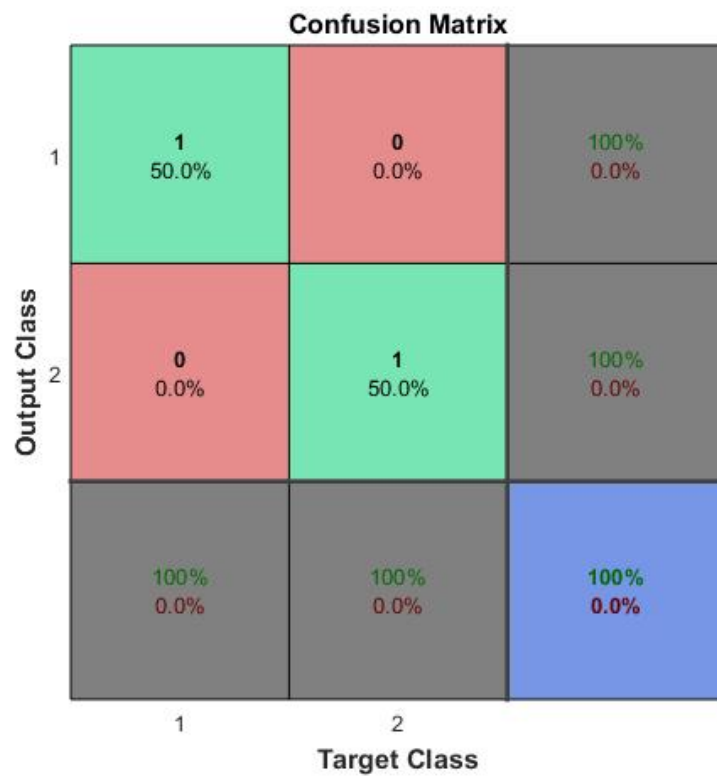


Figure 1: Confusion Matrix Generated During Evaluation

```

Training PatternNet...
Training FeedforwardNet...
Evaluating PatternNet:
Accuracy : 100.00%
Precision : 100.00%
Recall : 100.00%
F1 Score : 100.00%
Evaluating FeedforwardNet:
Accuracy : 100.00%
Precision : 100.00%
Recall : 100.00%
F1 Score : 100.00%
>>

```

Figure 2: Additional Output from Speech Feature Extraction or Training

Conclusion

This lab successfully implemented feature extraction and neural network classification for speech signals. MFCCs provided robust feature representation, while both PatternNet and FeedforwardNet achieved strong classification performance. The evaluation metrics and confusion matrix clearly show the effectiveness of the models for distinguishing between two speech samples.